

Problem

All of the values are in the range $[0, 100]$, so create an array of zeros, **result** = $[0, 0, 0, 0]$. The results of each iteration follow:

i	arr[i]	result
0	1	[0, 1, 0, 0]
1	1	[0, 2, 0, 0]
2	3	[0, 2, 0, 1]
3	2	[0, 2, 1, 1]
4	1	[0, 3, 1, 1]

Submissions

The frequency array is $[0, 3, 1, 1]$. These values can be used to create the sorted array as well: **sorted** = $[1, 1, 1, 2, 3]$.

Note

For this exercise, always return a frequency array with 100 elements. The example above shows only the first 4 elements, the remainder being zeros.

Challenge

Given a list of integers, count and return the number of times each value appears as an array of integers.

Function Description

Complete the countingSort function in the editor below.

countingSort has the following parameter(s):

- arr[n]: an array of integers

Returns

- int[100]: a frequency array

Input Format

The first line contains an integer **n**, the number of items in **arr**.

Each of the next **n** lines contains an integer **arr[i]** where $0 \leq i < n$.

Constraints

$$100 \leq n \leq 10^6$$

$$0 \leq arr[i] < 100$$

Sample Input

```
100
63 25 73 1 98 73 56 84 86 57 16 83 8 25 81 5
```

Sample Output

```
0 2 0 2 0 0 1 0 1 2 1 0 1 1 0 0 2 0 1 0 1 2
```

Explanation

Each of the resulting values **result[i]** represents the number of times **i** appeared in **arr**.

Change Theme

Language

Python 3



```
1  #!/bin/python3
2
3  import math
4  import os
5  import random
6  import re
7  import sys
8
9  #
10 # Complete the 'countingSort' function below.
11 #
12 # The function is expected to return an INTEGER_
13 # The function accepts INTEGER_ARRAY arr as para
14 #
15
16 def countingSort(arr):
17     result = [0] * 100
18
19     for num in arr:
20         result[num] += 1
21
22     return result
23
24 if __name__ == '__main__':
25     fptr = open(os.environ['OUTPUT_PATH'], 'w')
```

Line: 36 Col: 17

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Run Code

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☐ Test against custom input

Test case 0

Compiler Message

Test case 1

Success

Test case 2

Test case 3

Test case 4

Test case 5